Zhiyuan Holly Zhang

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Ph.D. student in Computer Science at UCLA focused on LLMs, retrieval-augmented generation (RAG), privacy-preserving machine learning, and ML systems. Seeking Summer 2027 research, machine learning, and software engineering internships.

EDUCATION

University of California, Los Angeles (UCLA) **09/2025-Present (expected 06/2030)**
Ph.D. student in Computer Science

The Hong Kong Polytechnic University (PolyU) **08/2021-06/2025**
BSc (Hons) Scheme in Computer Science

- **Awards:** Dean's Honours List of Outstanding Student in the 2022/23 Academic Year; Best GBA Solution in Cathay Hackathon 2023; Wong Tit-shing Scholarship

SELECTED RESEARCH & INDUSTRY PROJECTS

Private-RAG: Answering Multiple Queries with LLMs while Keeping Your Data Private **02/2025-12/2025**
Research Assistant, Supervisor: Dr. Ruihan Wu, University of California, San Diego

- Developed a document-level differential privacy approach for multi-query RAG, achieving $\varepsilon \approx 10$ across 100–400 queries, compared with $\varepsilon > 1{,}000$ under conventional sequential composition.
- Built and evaluated DPR-based RAG pipelines with OPT-1.3B, Pythia-1.4B, and Mistral-7B across Natural Questions, TriviaQA, MQuAKE, and ChatDoctor, covering ~20M Wikipedia documents and 200K healthcare QA pairs.
- Improved robustness against multi-query membership inference attacks, reducing attack AUC from ~0.60 to 0.50.

Wrong-Way Pedestrian Detection Using Computer Vision **06/2025-09/2025**
Computer Vision Engineer, Supervisor: Mr. Zhihui Lin, Xin Da Tong Co.

- Developed a real-time YOLOv8-based computer vision system for wrong-way pedestrian detection across escalators in 8 metro stations, serving 300K–500K passenger movements per day.
- Improved detection reliability through region-of-interest filtering, track-length thresholds, and temporal smoothing, reducing false alarms from stationary passengers, occlusion, and noisy detections by 16%, enabling more reliable 24/7 automated safety monitoring..
- Built end-to-end OpenCV video analytics pipeline covering object detection, trajectory tracking, wrong-way event logging, visualization, and alerts, reducing dependence on continuous manual CCTV monitoring.

Airport Wandering Tourist Detection Using Computer Vision **06/2024-09/2024**
Machine Learning Engineer, Supervisor: Mr. Yi Dai, Changsha Airport.

- Implemented passenger detection using YOLO and multi-object tracking with DeepSORT, serving 90K–100K passengers per day and over 10M passenger journeys during the project period.
- Designed trajectory-based behavioral features, including direction, dwell time, repeated paths, and spatial displacement, to distinguish normal passenger flow from potential wandering behavior.

PUBLICATIONS

1. R. Wu, E. Wang, **Z. Zhang**, Y. Wang. Private-RAG: Answering Multiple Queries with LLMs while Keeping Your Data Private. Accepted at the Conference on Language Modeling (COLM), 2026.
2. Y. Liu, Y. Sang, X. Liao, M. Sun, B. Yang, **Z. Zhang**, C. Stanford, H. Ma, J. Ma. GPS-Enhanced Tourist Mobility Modeling with Seasonal Spatial Priors and LLM-Based Activity Chain Generation. Accepted at IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), 2026.
3. T. Koga, R. Wu, **Z. Zhang**, K. Chaudhuri. Privacy-preserving retrieval-augmented generation with differential privacy. arXiv:2412.04697.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL, R, JavaScript, C
ML/AI: Transformers, Diffusion Models, Diffusion Transformers (DiT), Neural Operators, PyTorch, TensorFlow
LLM & Retrieval: Retrieval-Augmented Generation (RAG), Dense Passage Retrieval (DPR), LLM Evaluation
Model Optimization & Deployment: 4-bit Quantization (Hugging Face), ONNX, TensorRT